DunesForge - QuickStart from your first scene to a finished shot

An expanded working guide to sand, dunes, snow and water in Blender.

DunesForge 0.85 beta | Full English edition | Local documentation, available offline.

New here? Start with Getting Started (ten minutes). Something not working? See Troubleshooting.

How to use this guide

This QuickStart takes you through the decisions involved in building a scene. It explains what to click, why you are doing it, what should change and how to judge whether the result makes sense. You do not need to read the entire document before trying the add-on. Start with **Your first scene**, then choose an exercise that matches your project.

To work with your own landscape, follow **Import a landscape heightmap**, **Paint your starting sand** and **Wind direction** in that order. To explore a ready-made example, begin with a demo and read the chapter about its main process: snow, water, masks or vegetation. Before delivering a shot, read saving and export. Those chapters explain why a scene file, a checkpoint and a bake serve different purposes.

Control names in bold retain the wording used in the English interface, so you can find the right button without guessing a translation. When a term first appears, the surrounding text explains its purpose. Changing the documentation language does not change the language of Blender's buttons.

Each exercise provides a goal, preparation, steps and a checkpoint. The checkpoint matters: if your result differs from the description, stop and inspect the listed settings. Adding more features usually makes the cause harder to identify.

- Installation and updating - if the add-on is not yet available in Blender.
- Complete feature guide - for an explanation of a particular control.
- Demo gallery - to choose a type of landscape before starting.

The illustrations show actual built-in demos. They demonstrate a kind of effect, rather than promising an identical image after each exercise. Framing, lighting, resolution and the number of calculated steps all influence the result.

What you are building in DunesForge

Think of a scene as several cooperating layers. **Terrain** defines the area and its shape. **Bedrock** defines the floor below loose material. **Sand** lies above it and can be transported. **Wind and other processes** change the distribution of material. The **surface shader** shows color, fine grain, moisture and other surface characteristics.

For example, suppose you are preparing a rocky valley and want sand only at its bottom. The valley should remain the underlying ground. You add starting sand at the bottom, and wind reshapes that layer. If you interpreted the whole valley as loose sand, the simulation could also change mountains that were supposed to remain rock. This is a decision about what the scene consists of, with consequences beyond its appearance.

Another example is fine surface rippling. It may come from the calculated shape, or from the material and rendered detail. These can look similar at a distance but need not produce the same geometry on export. Evaluate the large shape, simulation behavior and surface appearance separately.

DunesForge uses a regular heightfield: each location on the ground plane has one height. This suits dunes, valleys, slopes, shores and snowdrifts. A cave or overhanging ledge would require several heights at the same location. Such geometry is not ordinary DunesForge simulation terrain. Additional objects can exist in the scene without changing how the terrain itself works.

The essential distinction: a terrain heightmap describes the landscape's shape. A sand map describes the thickness of sand above the ground. A mask describes where and how strongly a

particular feature acts. Three similar black-and-white images can therefore have three entirely different jobs.

Prepare Blender and a comfortable workspace

1. Save your current project and open a new scene for learning. Demos create their own objects, lighting, camera and materials; an empty scene makes their contents easier to identify.
2. Place the pointer over the 3D Viewport, press **N** and select the **DunesForge** tab. If it is missing, check that the add-on is enabled in Preferences. After an update, restart Blender and read the version number in the panel.
3. Widen the sidebar until you can read the complete labels. A narrow panel can shorten names and make similar controls harder to distinguish.
4. Keep **Simple** for the first exercise. Switch to Standard or Advanced when the instructions lead to a control you need.
5. Initially inspect the shape in a simple display mode. Use material or rendered preview when evaluating color, wet areas, snow or fine detail. Gray terrain in Solid mode does not by itself indicate a missing material.
6. Read the GPU or CPU status. A selected option and acceleration actually running are not always the same. On CPU, begin with a small grid; start at a manageable resolution while learning.

The first panel contains **QuickStart** and **All Features**. A panel's help icon opens the explanation of that group. Right-click a setting and choose **Explain this DunesForge option** to open that particular control's description. Choose the guide language with **EN** / **PL** next to **Guides**. The manuals are PDF files in the add-on's docs subfolder; use your PDF viewer's search (Ctrl+F) to find a word. The help icons open the exact page in Adobe Acrobat or Reader, SumatraPDF and browser PDF viewers; other viewers open the guide at its first page.

Organize your project folder around the .blend file, input maps and exported results. Use new names for successive trials, such as valley_01 and valley_02. For longer calculations, keep a short

note of the demo or preset, resolution, seed and significant change. After several variants, that note is more reliable than remembering what you changed.

Your first scene: learn the full workflow

Goal: see a working scene, change one parameter and keep your own variant. You are not yet building everything from scratch.

1. Find **Start with a Demo or Preset** in the main panel. Select **Desert: Dunes + Obstacles** and click **Create Demo Scene**.

2. Wait for creation to finish. A demo may perform warm-up: a series of calculations before displaying the scene. This avoids beginning with completely flat sand. Do not click again simply because the view has not changed immediately.

3. Inspect the terrain from above and at a low angle. From above, ridge direction is easier to read; from the side, height and slopes are clearer. Find obstacles and identify the sheltered regions around them.

4. Enable Material Preview or Rendered display. If the scene looks different from the gallery, check camera and lighting too. Another viewing angle can hide small forms without any simulation change.

5. Start **Play / Pause** in DunesForge. Watch for a short interval, then stop. Follow a particular ridge or a patch behind a rock. Judging the whole landscape at once makes slow movement harder to notice.

6. Save under a new name. Note the wind direction, change only **Wind Direction**, run another interval and compare the same area.

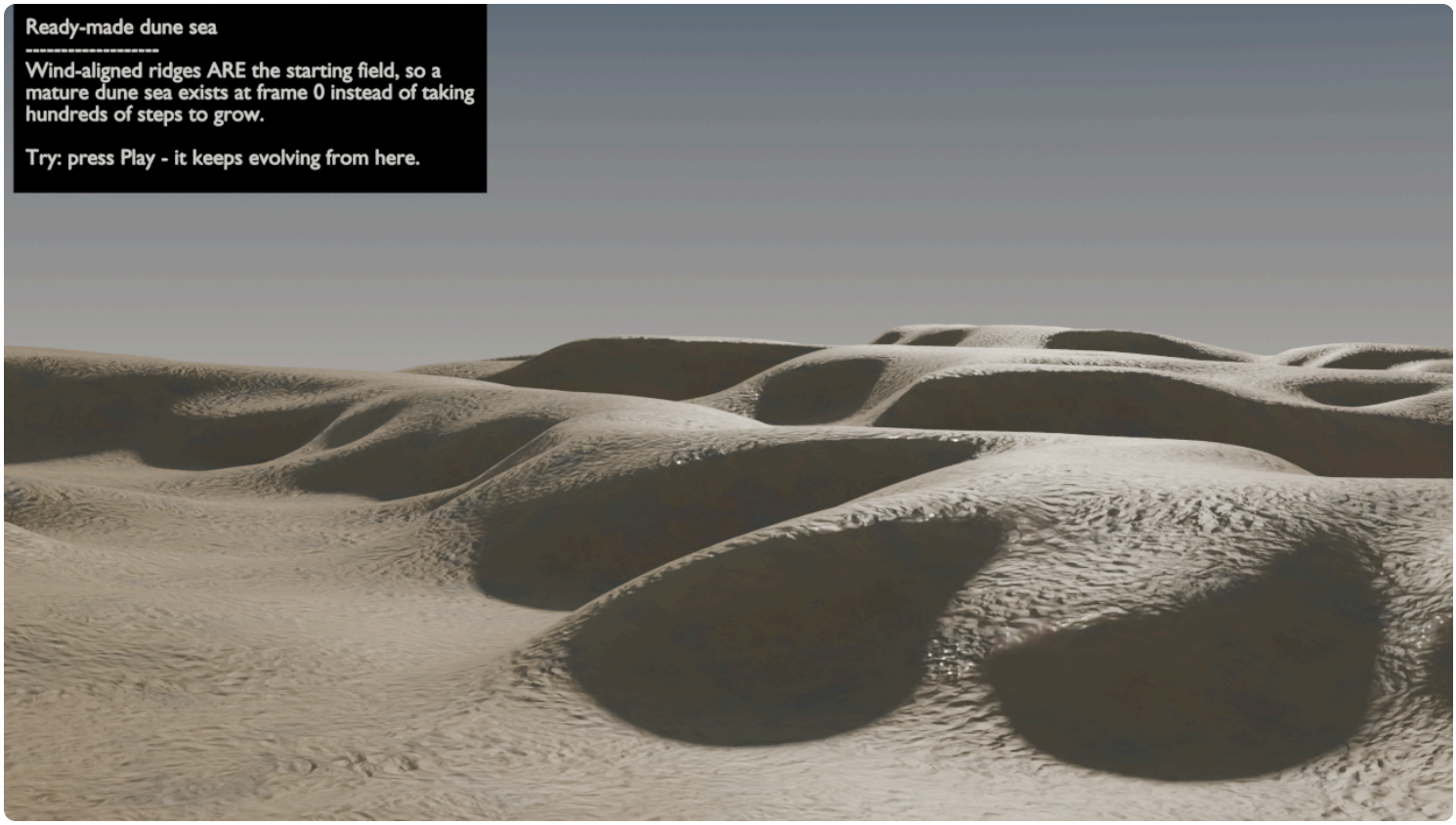
7. Do not expect dunes to rotate immediately. Wind changes where sand is removed and deposited; the existing shape needs time to rebuild.

8. Stop at a result you like and save the .blend. If you need to return to the calculated state, also follow the checkpoint chapter. Saving the scene does not replace every kind of simulation-

state storage.

Checkpoint: the terrain is visible, calculation starts and stops, and changing the wind affects later development. If nothing moves, first check that loose sand exists, the simulation is initialized and **Freeze Dunes** is off.

Small comparison exercise: capture two views of the same place with an unchanged camera. Between them, perform only another series of steps. This helps distinguish actual terrain movement from a shadow change caused by rotating the view.



Example from TERR_DUNEFIELD. Observe the ridge paths and the ratio between the height and width of the forms. Material helps reveal them but does not replace the surface shape.

Simple, Standard and Advanced: when to switch

Simple covers the complete basic workflow: terrain selection, import, starting sand, painting, and the main process and appearance controls. It uses the same simulation with a smaller set of visible controls.

Choose **Standard** when building relationships yourself: refining water, vegetation, materials, erosion by objects, dust devils, recording and export. A concrete question is a good reason to switch, such as: how do I record the dunes as a frame sequence?

Advanced provides full control, including accuracy, detailed models, performance and diagnostics. You do not need to use every available slider. Some concern particular models or experiments and are irrelevant to an ordinary dune field.

Switching UI levels does not disable functions you previously enabled. If you activate an extra process in Advanced, returning to Simple may hide its controls while leaving the process active. Check the list of active features outside the current interface level so you can distinguish a hidden setting from a disabled feature.

Some controls appear only for the selected operating mode. **Target Value**, for example, is relevant to the Set brush; image parameters appear for an Image stamp. Their absence in another mode is not necessarily a problem. Check the selected model, brush target and required inputs before changing UI level.

Understand scale before increasing quality

Three decisions are easily confused: area size, calculation density and dune size. The size of the terrain object describes the extent of the landscape (**Create Terrain** makes a 200 m plane; scale the object to change it). **Solver (sim grid)** describes how many samples are used to calculate it. **Dune Scale** helps control the scale of forms within that area. Each has a separate role.

For perspective, the default 200-meter terrain with 128 samples per side has spacing of about 1.6 meters. This is a planning approximation, not a separate solver calibration. A track a few centimeters wide will not become readable on that grid simply by increasing its depth. It must occupy a meaningful part of the grid.

Doubling resolution along each side gives roughly four times as many samples. Memory, computation and geometry update costs rise. Try 128, or another modest available value, first; test higher resolution after establishing the composition. This is a learning starting point, not a mandatory final quality setting.

Render Subdivision increases displayed geometry through interpolation. It can smooth visible edges, but it does not add new detail to the calculated movement of sand. Fine material grain likewise does not increase simulation resolution.

When importing, begin with **Height Scale = 1** and set elevation through **Image Amplitude**. A very large amplitude compensated by a very small display scale is not a straightforward substitute for a physically gentle slope. If sand continually slides off the entire terrain, check steepness and proportions before adjusting the angle of repose.

Tutorials: choose your route

You can follow these exercises separately, but they are most useful in this order: terrain, starting sand, wind, obstacles, appearance and saving. Snow, water and vegetation extend the same approach. In each new project, get one process working before adding the next.

Save a variant after a successful stage. If the next experiment fails, you can return to a working point instead of reconstructing forgotten settings. Checking the result means inspecting the actual scene, not merely enabling the named checkbox.

Tutorial 1. Import your own landscape heightmap

Goal: preserve your terrain's shape and prepare it for sand. **Prepare:** a heightmap and an empty scene. Begin with a smooth, moderately sloped landscape containing an easily recognized valley and hill.

A heightmap should actually describe elevation. A color photograph of a desert records light and color, so its brightness is not reliable elevation data. For a large height range, higher precision such as EXR or a suitable 16-bit map is preferable. Too few levels can produce terraces and steps.

1. Click **Create Terrain**. Set **Solver (sim grid)** before importing. Resolution determines how accurately the shape can be sampled.

2. Select the file in **Load Heightmap Bitmap**. Set **Height Scale = 1** and choose **Image Amplitude** for the intended elevation range. Use moderate relief for learning. For a map normalized from 0 to 1, amplitude 10 corresponds to approximately 10 height units before additional transformations.

3. Initially leave **Image Offset** at 0. It shifts the whole terrain vertically rather than increasing the difference between valley and summit. Check the range in EXR/TXT data: not every file contains only values between 0 and 1.

4. Click **Load Heightmap Image**. Inspect before simulating. Locate a known feature to check orientation and proportions.

5. Check **Start From = Current Mesh**. This means starting from the mesh's present shape. Selecting a noise generator would prepare a different initial shape.

6. Leave **Terrain itself is loose sand** off if the landscape is the underlying ground. Enabling it changes the input's interpretation: the imported shape becomes loose material that can be reshaped.

7. If the image is not tileable, check **Keep opposite edges separate / Edge Clamp**. Opposite borders of a mountain map usually do not match. Separating them helps avoid treating them as one continuous landscape, although it cannot correct every source-map problem.

8. Save an imported-terrain .blend variant. Add sand using the following exercises. Use **Initialize Simulation** after preparing that starting layer.

Checkpoint: after Initialize, you recognize the same valley and hills. Sand may cover fine relief, but the ground should not be replaced by random noise. If it changes, check Current Mesh and Terrain = Sand.

If you see spikes: inspect the map and the tools for removing isolated bad samples in the more detailed UI. Avoid smoothing the entire terrain excessively when the problem is a few bright pixels.
If you see steps: check image precision and amplitude. Filtering can help but cannot recover real data absent from the file.

Comparison: import the same map into a copy at half the amplitude. Judge slopes and room for sand. Choose the landscape proportions before a long simulation rather than attempting to

compensate with every physics parameter.

More on inputs: terrain and sand, Image Amplitude.

Tutorial 2. Your own mesh and terrain from scratch

Goal: choose the right route for existing geometry. A large vertex count alone is insufficient: DunesForge needs a regular square grid that can be read as a heightfield.

The most reliable route to a hand-shaped start is **Create Terrain**, preserving its grid layout while changing heights. Adding arbitrary vertices, deleting sections, changing topology or folding the surface into overhangs breaks this simple heightfield structure. For an irregular model, a heightmap can be a more convenient intermediate input.

1. Select the correct mesh in **Terrain**. Do not assume selecting an object in the viewport assigns it as the add-on's terrain.
2. Click **Check Terrain** (main panel, below the simulation buttons). Address the reported requirements before adding masks and materials.
3. Use **Current Mesh** to keep the shape. Leave **Terrain itself is loose sand** off for a persistent landscape.
4. Prepare starting sand, initialize and check the shape. Changing resolution or rebuilding the grid can affect detail, so preserve the original.

Starting from scratch, choose a generator in **Start From** after Create Terrain. Noise supplies a landscape base; dune variants prepare particular initial forms. For a generated landscape that should remain rock, deliberately use **Protect generated terrain as rock**. Otherwise the generated hills may be interpreted as loose material.

Checkpoint: you know whether the simulation should reshape the entire landform or only sand on its surface. If uncertain, revisit the ground settings before running it.

Tutorial 3. A uniform layer and sand in valleys

Goal: compare three starting approaches: full coverage, rule-based placement and hand painting. Rules suit similar behavior across large areas; painting controls individual places.

For an ordinary start, **Even sand layer (start)** sets loose-layer thickness. Enabling **Starting Sand from a Map** gives that role to the image or painted layer. Do not judge Initial Sand Depth while another starting-sand source is active.

1. Use prepared terrain with Current Mesh. Turn Terrain = Sand off. For the first comparison, also disable Starting Sand from a Map.

2. Choose full coverage in **Sand Placement**. Use a modest layer that does not hide all terrain relief. Initialize and inspect briefly to distinguish sand from the ground.

3. Save the result and return to preparation in a copy. Choose a height restriction for low-lying sand. Placement height values are percentages of the terrain range: 25 means 25%, not 25 meters or 0.25%.

4. To avoid steep walls, use a slope restriction. These values are in degrees. Begin with an obvious broad group of slopes, then narrow the range.

5. **Combined Filters** can combine height, slope, depressions and irregular patches. Add conditions individually. Requiring all conditions can leave only a small eligible part of the terrain.

6. If the result is empty, disable the last condition or loosen its threshold. **Combine with OR** expands the allowed region to areas meeting any condition. It is not simply a gentler version of the same filter.

Checkpoint: the initial distribution matches your intended region. Inspect immediately after Initialize, before wind changes it. If nothing appears, temporarily broaden the restrictions and check the source rather than increasing unrelated process settings.

Initial placement need not hold sand in that region throughout the animation. **Place Hold** limits later movement outside the starting region. Begin without strong holding to understand natural

movement, then decide whether the composition needs this additional control.

Tutorial 4. Paint your own starting sand

Goal: prepare separate sand patches on your landscape, then let wind reshape them. **Prepare:** working terrain, preferably gently sloped, before Initialize. Begin with large, recognizable patches.

Set the correct sand source

1. Set **Start From = Current Mesh**. Turn **Terrain itself is loose sand** off when painting above a fixed landscape.
2. Enable **Starting Sand from a Map**. This includes the painted layer; you do not need an externally prepared image.
3. Leave **Sand Map** empty. An assigned image takes precedence over painting. The cross beside the field removes that assignment; it is different from painting the image black.
4. Keep placement simple for this first exercise, without extra restrictive filters. Otherwise it is harder to distinguish a brush problem from a later starting-condition restriction.

Paint and inspect the layer

1. Choose **Paint = Sand Height**. Before Initialize, this prepares starting sand; after Initialize, it edits the current simulation surface.
2. Choose **Mode = Set** and a moderate **Target Value**, such as 1 with Height Scale = 1 for a test. Set moves toward the target beneath the brush; a quick, weak touch may not reach it immediately.
3. Choose a **Radius** large enough to cover a visible area rather than a single point. **Strength** controls stroke intensity. Begin moderately and make several deliberate strokes while watching the preview.

4. Click **Paint Starting Sand**. Paint with the left mouse button in the 3D Viewport. The wheel changes radius. **Shift** reverses the action according to the mode. **Esc** ends painting; do not treat it as a guaranteed undo of the entire stroke.

5. Paint a broad patch on the valley floor and a smaller one nearby, leaving a clear gap. A soft border is more natural than a sudden vertical edge on a thick layer.

6. Finish and inspect the preview. Before Initialize, the painted start is stored as mesh data. The colored preview shows its coverage; geometry may not yet rise in those places.

Turn preparation into a running simulation

1. Save a sand-before-initialization variant. Click **Initialize Simulation** and inspect from a low angle.

2. Confirm sand lies above the floor in the painted areas, then start Play. Wind can move material beyond the original patch and connect areas: the painted start is not a fence.

3. Stop and save another variant to compare the initial and developed states.

Checkpoint: before Initialize you see painting data, after Initialize you see the corresponding loose layer, and during calculation its shape can change. Inspect these three stages separately.

Add or Set? Add gradually builds up material; repeated strokes can increase the amount. Set approaches a chosen value. Smooth softens transitions. Choose the mode for the task instead of attempting everything with an increasingly strong Add brush.

If nothing appears after Initialize: check Starting Sand from a Map, the empty Sand Map field, correct terrain and stored painting. Confirm you did not paint a different mesh from the one assigned in Terrain. If sand appears everywhere, check the active source and whether you are looking at an earlier initialized state.

Fill, Clear and Reset affect the whole selected field, not just the cursor area. Clear writes zero; removing a mask removes its presence. Fields with other defaults or automatic input can behave differently after these operations. Always check the Paint target first.

Details: painting, starting sand map.

Tutorial 5. Sand from an image and combining layers

Goal: use a sand map prepared outside Blender. This differs from terrain-height import: here the image describes how much loose sand lies above existing ground.

1. Prepare and check the terrain. Enable Starting Sand from a Map and assign an image in **Sand Map**.
2. Understand the input: in a simple reading, black represents no thickness and brighter areas represent more sand. **Sand Map Depth (m)** sets the thickness assigned to white. Identical-looking images can produce different layers if this value changes.
3. In Simple or Standard, check map fitting and preview. Inspect whether the pattern lies where intended, especially if the image and terrain have different proportions or the map was prepared for another wind direction. Rotating, flipping or offsetting the map needs **Advanced** (box **Edit the map**).
4. Choose **Replace** for straightforward reproduction of the supplied thickness. When combining with an existing layer, understand the mode before applying it.
5. **Max (keep the deeper)** retains the deeper layer, so a dark area in the new map may not remove existing sand. **Add** adds material. **Subtract** removes it. **Blend** mixes according to its factor. Modes using existing thickness need a ground reference against which to evaluate it.
6. Make a small test and inspect it before expanding the scene. A separate preview object shows the predicted shape; it does not automatically become another active simulation terrain.

Checkpoint: changing the depth assigned to white produces an understandable change in sand amount, and dark areas behave according to the combination mode. To remove sand through image painting, do not automatically choose Max: its purpose is to preserve the thicker layer.

Tutorial 6. Wind direction and dune scale

Goal: guide forms without randomly changing physics. Begin with **Terrain: Ready Dune Field** or a working desert demo. Avoid changing terrain size, resolution and preset simultaneously.

1. Choose an area showing several ridges. Save a variant and read **Wind Direction**.
2. Run briefly with the current direction and stop. Change it substantially, for example by about a quarter turn, and observe the same area.
3. Watch gradual rebuilding: which slopes become exposed, where material accumulates and whether old ridges act as obstacles to the new pattern.
4. Return to the saved point and change only **Dune Scale**. Compare proportions. To evaluate starting settings under identical conditions, use the same saved start rather than an increasingly mature landscape.
5. Only then add modest direction variation, such as **Wind Wobble**. Strong variation can blur the dominant direction and the clear shelter behind an obstacle. Establish orderly ridges with simpler wind first.
6. For phased wind, inspect **Phase Steps**. It appears when **Wind Mode** is **Linear / Seif** or **Star**. This is the duration of a direction in solver steps. Short phases mix influences more frequently; long phases allow one direction more time to reshape the surface.

Steps/Frame controls steps per playback frame. A higher value advances more evolution along the timeline and costs more per frame. It is not equivalent to stronger wind. **Run Steps** calculates a series without continuous preview refresh, which is useful for developing the terrain itself.

For moving objects, gates, animated sources or additional effects, also play the frames. A solver-step series does not replace all Blender timeline events. If a demo depends on a rising sluice gate (an experimental shallow-water demo), check its motion as well as the water state at a single frame.

Checkpoint: you can independently adjust direction, form size and playback progression. When an effect appears faster, you can tell whether this comes from steps per frame or different material

behavior.

Tutorial 7. Rock, wind shelter and protected terrain

Goal: distinguish an object that changes wind flow from a mask limiting material changes. Begin with **Showcase: What Objects Tell The Sand**, then try one object on simple terrain.

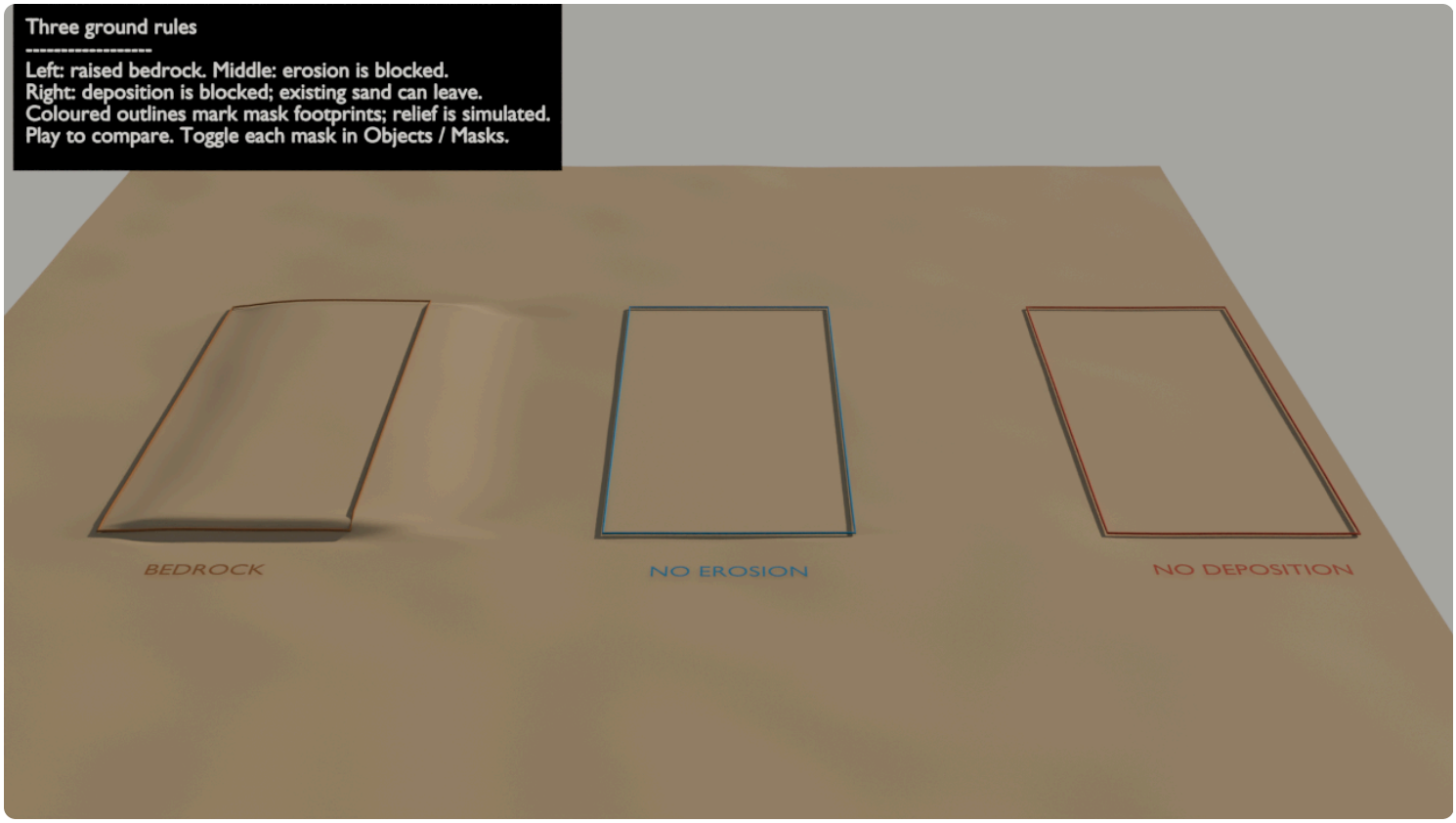
A rendered rock does not automatically become a simulation obstacle. The add-on needs its intended role assigned. One object can feed several mechanisms, but those mechanisms are not equivalent.

1. Open Showcase: What Objects Tell The Sand and inspect the marked regions. Compare ground, erosion protection and restricted deposition. Run briefly to see behavior rather than just the initial picture.
2. In a copy of simple terrain, place one obvious object. In **Objects / Masks**, enable the relevant obstacle handling and assign the object or collection to **Obstacle**.
3. Check wind direction. Observe in front of the object, along its sides and behind it. Record a short interval with the effect active.
4. In a comparable copy, disable the obstacle contribution while preserving the visible object and the same starting state. If both look identical, check assignment and whether mobile sand exists in the observed area.
5. Use **Erodibility** to protect an area against sand removal. Zero means no susceptibility to erosion. It does not imply that the area cannot receive sand from elsewhere.
6. Use **No-Deposit** to restrict accumulation. Existing sand may still be removed, gradually exposing the area.
7. Treat **Bedrock** as the underlying ground: it defines the lower boundary of loose material. It is not another name for a general no-change mask.

Checkpoint: you can predict which area is sheltered, which should not release sand and which should not receive it. If all three behaviors are needed together, configure them deliberately

instead of expecting one field to supply everything.

Painted masks and their previews help define precise boundaries. Inspect coverage before judging strength. A strong parameter cannot correct a mask located somewhere other than the rock.



Showcase: What Objects Tell The Sand compares mechanisms. Notice the different roles of the three regions, not only their colors in one frame.

Tutorial 8. Painted masks and stamps

Goal: use the brush deliberately rather than always treating it as terrain raising. In **Interactive Brush**, choose **Paint**, the edited field; **Stamp**, the stroke shape; and **Mode**, how values change.

1. Choose one target, such as erosion protection or vegetation. Read its value meanings: 0-1 suits many masks, while sand height and snow depth represent different quantities.
2. Use modest strength, a reasonably large radius and a soft edge. Enable the available field preview, paint one patch, finish and confirm that you edited the intended terrain.

3. Run the relevant feature and inspect the result. Painting values does not supply every prerequisite of a model. Avalanche weakening, for example, needs snow and a slope; there is nothing to release on an empty flat surface.

4. Add denser patterns after this test. If you overdo it, restore the saved variant or use undo tools appropriate to the current DunesForge state.

5. When painting after Initialize and retaining supported masks for another start, check **Save This Mask** / **Save All** under **Keep for next run**. Test reinitialization in a copy. Retaining a mask is different from saving the entire simulation history.

Barchan Stamp places a crescent-dune seed. **Dome Stamp** supplies a rounded form and **Ridge Stamp** a crest. Treat these as starting shapes that wind continues to change. **Scatter Clumps** distributes smaller patches, useful for irregular clusters. **Image Stamp** uses image brightness. Not every stamp makes equal sense for every brush target.

Checkpoint: before pressing Fill or Clear, you can name the field that will change across the terrain. If you meant to edit sand but Paint still targets vegetation, you will edit vegetation regardless of the brush tip's appearance.

Tutorial 9. Snowdrifts beside an obstacle

Goal: distinguish a white material from a snow layer that actually changes. Start with **Snow: Drifts + Fence**. The larger **Grand Scene: Winter Storm** scene is the next step after understanding the basic example.

1. Create the demo and inspect the fence and ground on both sides. Notice wind direction as well as camera position.

2. Save before experimenting. Run briefly and observe where a drift grows. Keep a fixed view to compare shape changes.

3. In **Snow**, change only **Snowfall**. Precipitation supplies fresh material and affects snow amount over time. Compare variants from the same starting point over similar step counts.

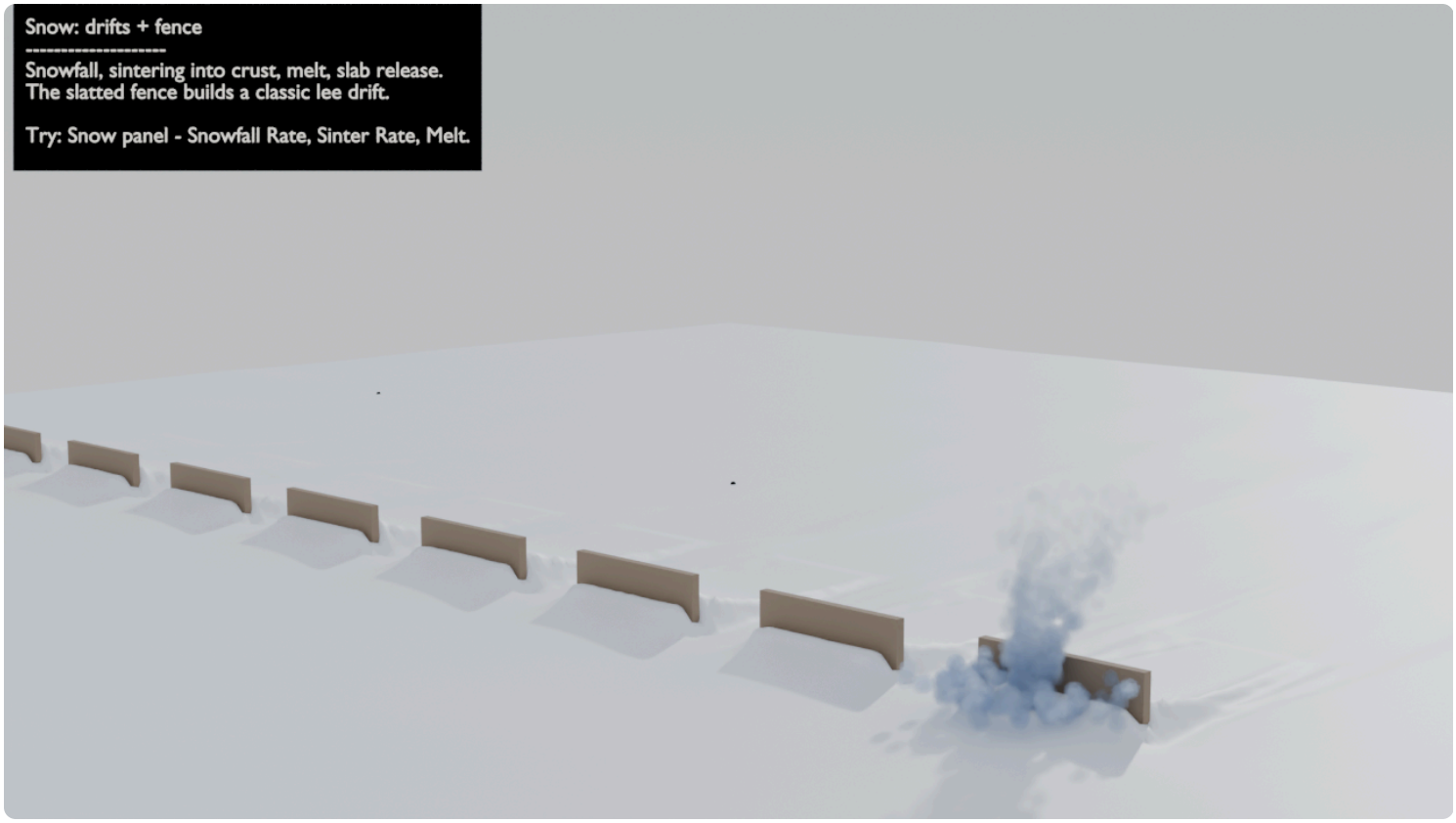
4. In a separate trial, change **Sinter**, the strengthening and bonding of the layer. Judge later snow behavior rather than expecting an immediate material-color change.

5. Inspect **Melt** and temperature settings if using the temperature model. With several melting mechanisms active, setting one slider to zero may not stop all melting.

6. For painting, distinguish **Snow Depth** from **Snow Crust**: fresh snow amount and crust are different layers.

Checkpoint: snow changes shape or coverage, and the material reveals that change. If everything stays white during melting, determine whether the material responds to the snow field or simply stays white.

For avalanches, choose a dedicated avalanche demo. Suitable slopes and material are required; enabling avalanches does not create mountains or add snow. Study working conditions in the example before reproducing them yourself. Avoid changing default physics solely to force an effect on terrain that is too flat.



In Snow: Drifts + Fence you can evaluate the drift and airborne particles separately. No visible particles in one frame does not prove that the snow layer is missing; these are different scene elements.

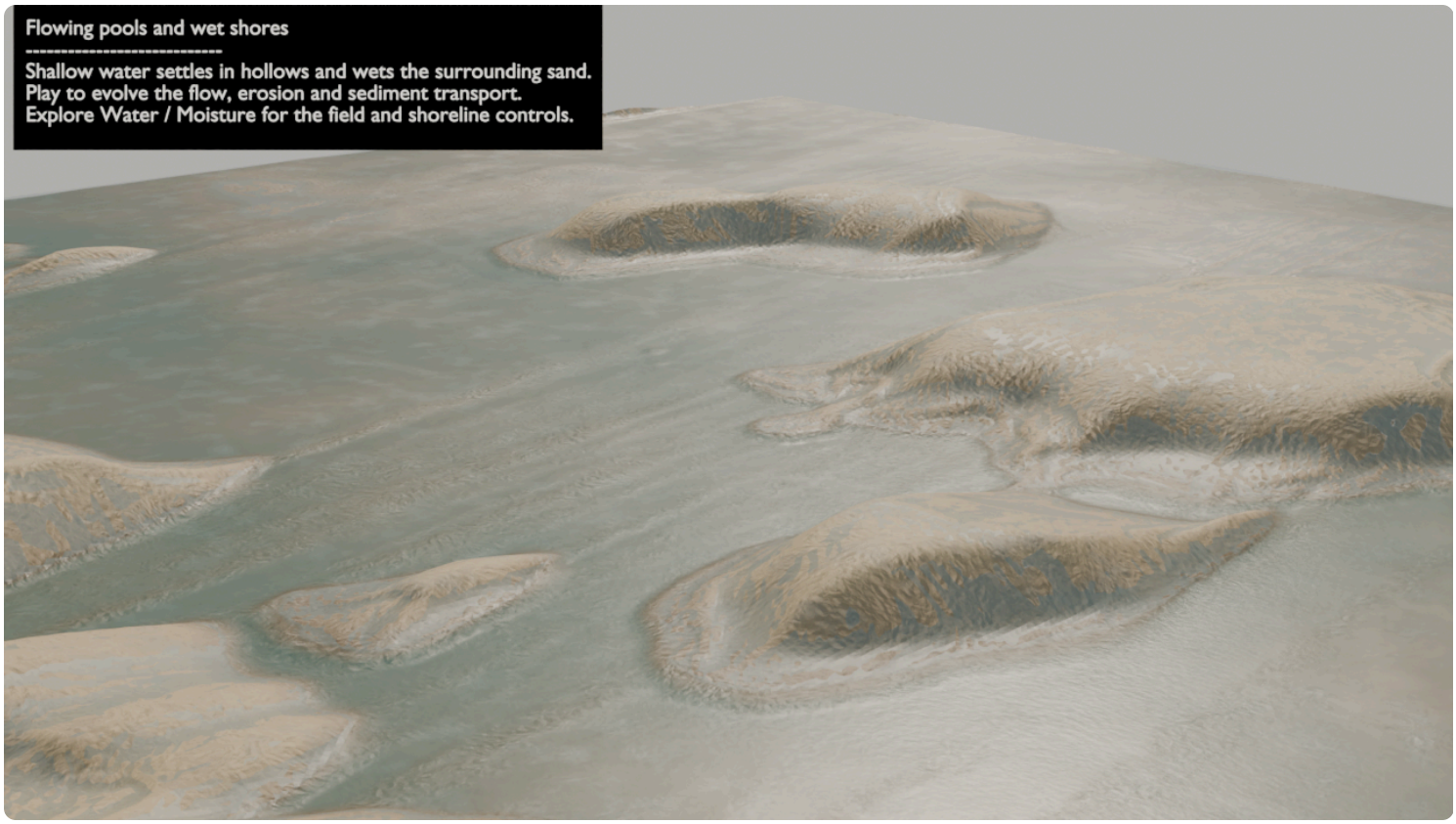
Tutorial 10. Water, wet sand and rain

Goal: distinguish the water surface, calculated flow and moisture in sand. Start with the simpler **Oasis: Water + Valley Masks** to understand a water-level workflow. **Water: Pools That Flow (Experimental)** previews the shallow-water model, which will probably arrive in a version after 1.0.

1. Create the demo and find water, shoreline and dry ground. Choose a camera where each occupies a visible part of the image.
2. In **Water / Moisture**, identify the active model. A simple water level suits a lake or flat shoreline. A shallow-water field (**Shallow-Water Simulation (Experimental)**) calculates local depth and flow across terrain.
3. Play and watch the wet region's boundary. A change in appearance need not mean a change in flow. Color, roughness and small waves also belong to the appearance layer.
4. When studying water's influence on sand, watch the land too. Ground hidden beneath opaque water is a poor place to judge subtle sand changes.
5. Open **Rain: Wet Sand + Plants** separately for rainfall. First observe moisture buildup and appearance, then compare sand movement. Combine rain with standing water and vegetation afterwards.
6. Choose a material variant suitable for the scene. Calculated moisture may remain hard to see if the material does not read the relevant fields.

Checkpoint: you can identify whether a change concerns water amount, motion, sand moisture or appearance alone. If everything seems still, check whether equilibrium has been reached. A calm lake need not continuously show an obvious current.

The shallow-water field describes surface flow above terrain. It does not represent an arbitrary liquid volume with caves, overhangs and fully three-dimensional splashes. For a sluice or moving gate, use the corresponding demo and play its object animation; Run Steps also follows its object animation. Shallow water remains experimental.



Water: Pools That Flow (Experimental) example. Study the shoreline and transition from wet to dry sand. Grand Scene: Dynamic Coast (Experimental) later provides a larger setting for related processes.

Tutorial 11. Vegetation that influences sand

Goal: prepare vegetation-stabilized areas and distinguish a coverage field from plant models. Begin with **Vegetation: Nabkha Hummocks** or **Plants: Valleys Only (Slope and Height Windows)**, the vegetation-in-valleys example.

1. Inspect vegetation distribution in the demo. Look at material color and sand shape around covered regions.

2. In **Vegetation**, identify the initial distribution source and the rules for subsequent growth. Where vegetation starts and where it can persist are separate decisions.

3. On a copy of simpler terrain, paint one vegetation patch. Begin with a large readable region rather than fine random noise.

4. Run briefly and compare sand movement inside and outside the patch. Keep the wind and available sand layer comparable so the difference can be attributed to vegetation.

5. Add height or slope restrictions later. If vegetation disappears, also inspect burial, moisture and growth conditions rather than only increasing initial paint strength.

Checkpoint: field distribution matches your composition and has an understandable effect on sand. Green shading need not mean individual grass blades. Material and model placement are separate ways of displaying the field.

Grand Scene: Living Oasis combines water, wet shores and vegetation. Open it after the simpler exercise to distinguish elements that drive behavior from those used for appearance.

Tutorial 12. Material, detail and airborne sand

Goal: refine the image while keeping appearance and physics distinct. Begin with stopped geometry whose large forms already work. Fine textures rarely rescue an unsuccessful large shape.

1. Click **Add Surface Shader** to assign a material using the add-on's fields. Try **Add Fast Surface (cheap)** for lighter work. **Flat Colour (diagnostic)** can help inspect field distribution itself.

2. First adjust broad differences: dry sand, ground, wet areas and snow colors. Work on fine grain and ripples afterwards.

3. In Standard or Advanced, under **3. Surface appearance** in the main panel, pick a suitable variant and apply it. A dry scene may not need every rain block, and a fast preview may not require displacement. Full is the most extensive variant; test it in your renderer and target scene.

4. Judge detail in the final frame. Grain exaggerated to show in a wide shot may resemble gravel in a close-up. Realistically small features may occupy less than a pixel at a distance.

5. Compare a simple material and the target material with unchanged camera and lighting. If a problem disappears with the simpler material, investigate appearance before repairing

terrain.

6. For airborne sand, open **Showcase: Airborne + Dressing**. Inspect **Sand Haze** and **Dust Devils** (Standard) separately. Frame particles against a readable background. Invisibility can involve lighting, warm-up, available material or position, as well as particle count.

Checkpoint: you know which details belong to terrain shape, material and additional objects or effects. Before export, decide which should be included in the result.



Showcase: Airborne + Dressing demonstrates effects above the surface. Renderer visibility and transparency matter; a single still frame does not describe the entire system's behavior.

Tutorial 13. Save a result and return to it later

Goal: choose the right save operation and test restoration. Stop playback first. Several buttons containing Save or Bake are not interchangeable.

A **.blend file** preserves the scene, objects and settings according to Blender's saving behavior. Also account for external images and resources. A **Checkpoint** saves supported solver state. **Snapshot Terrain** creates an object for geometry comparison. **Bake** records a selected result or sequence. **Bake State as New Start** commits the current result as the basis for another start.

Save the scene and checkpoint

1. Stop simulation and save the .blend under a new name. Note the add-on version.
2. Open **Export / Bake** in any UI mode. In the box **Save / resume dunes and wind**, choose a file in **Checkpoint** and use **Save State**. Give comparison variants separate names.
3. In a project copy with correctly assigned terrain and prepared simulation, test **Load State**. Read the operator's message; selecting a file alone does not prove restoration succeeded.
4. Compare the surface and run a few steps. Check required masks and other scene elements too. A checkpoint does not promise to save every camera, texture, animated object and auxiliary system.

Snapshot and a new start

A snapshot is useful for placing two stages side by side and comparing shape, especially when selecting composition. It does not automatically replace a checkpoint for restoring supported calculated state.

Use **Bake State as New Start** when you deliberately want the current result to become the next baseline. Preserve the earlier variant. If rendered detail is included in the committed shape (the noise icon next to **Bake State as New Start**), it can become part of the next start's geometry. Repeating this without inspection can add successive layers of roughness.

Reset to Commit and **Reset to Original** refer to different baselines. Availability depends on retained history. Global **Reset Everything to Startup Values** restores add-on settings; it is not a pause control or a safe way to inspect a previous frame.

Checkpoint: before closing, you can open a copy and recover the result you need. A file's existence alone does not prove that it contains the data required for your next session.

Tutorial 14. Export and a short animation test

Goal: prepare repeatable output for rendering or further work. Begin with a short range, such as a few frames, and a small test file. Catch scale, path and data-type errors before creating a large sequence.

1. Open **Export / Bake**. Heightmap and mask export work in Simple; for a frame sequence (Bake Dir, Run ID, Record Evolution) switch to **Standard**. Choose output for the next task: shape, heightmap, field maps or a sequence. Select according to the destination rather than a format's apparent sophistication.

2. For a heightmap, check range and scale after reading it at the destination. For displacement, also verify the density and subdivision of the receiving surface.

3. For a sequence, set **Bake Dir** and a separate **Run ID** for the trial. Distinct run names keep frames from different variants apart.

4. Check the start, end and output resolution. Decide whether visual detail belongs in the output. Preview, solver and exported-map settings concern different stages.

5. When recording playback, check frame recording and, where field-dependent appearance is needed, which masks are stored. Geometry alone may not reproduce changing moisture, snow or vegetation.

6. Play the short recording using **Bake Viewer**. Check the selected frame, timeline following and preview-object position. The original and preview can overlap, making judgment difficult.

7. Open the result in a separate scene copy or destination tool. Compare orientation, elevation and detail before extending the range.

Pay attention to **Simulate While Rendering**: it can advance simulation while rendering frames. Rendering the same frame number again may then produce a different state. For a repeatable shot, deliberately choose a recorded run and verify playback instead of assuming live simulation behaves like an already calculated animation.

Checkpoint: you can return to a recorded frame and see its expected state without accidentally evolving the simulation further. Finally compare lighting and material too: a correct heightmap does not automatically store the scene's entire appearance.

Learn from the built-in demos

A demo is a prepared scene for exploring a feature. A preset is a group of settings. Creating a demo and applying a preset therefore have different effects. A preset can change many parameters together; variants described as replacing terrain can also change your prepared shape. Save before comparing.

Choose an example according to the question you want to answer:

- **Terrain: Ready Dune Field / Desert: Dunes + Obstacles:** dune scale, wind direction and a basic desert.
- **Showcase: What Objects Tell The Sand / Showcase: Object Masks:** ground, protection, shelter and sources.
- **Snow: Drifts + Fence:** drifts near an obstacle; avalanche demos show snow on slopes.
- **Oasis: Water + Valley Masks / Water: Pools That Flow (Experimental):** simple water level and calculated flow.
- **Rain: Wet Sand + Plants:** rain and moisture; **Vegetation: Nabkha Hummocks / Plants: Valleys Only (Slope and Height Windows):** vegetation distribution.
- **Showcase: Airborne + Dressing:** suspended particles and vortices; **Showcase: Surface Detail / Showcase: Surface Shader:** detail and materials.

The four large scenes, **Grand Scene: Desert Expedition**, **Grand Scene: Living Oasis**, **Grand Scene: Winter Storm** and **Grand Scene: Dynamic Coast (Experimental)**, represent 400-meter areas combining compatible processes. With many processes active together, they are not the clearest first test of each individual feature. Understand a simple example first, then examine its role in the larger landscape.

A practical learning method: create a demo, save variant A, change one thing and save B. Inspect the same area through the same camera. For behavioral comparisons, begin both at the same point. Changing five parameters after several hundred additional steps does not reveal which parameter caused a difference.

Read demo descriptions. Some features need motion, a slope, fresh snow, moisture or a particular animation time. Enabling a switch without the required conditions may show no visible result. Working examples help identify those inputs.

Technical route: five principles for organizing a scene

1. Separate preparation from live calculation

Start From, starting sand and some masks supply initial conditions. Wind direction and many process controls act on the running state. Other elements depend on scene frames. When diagnosing, first ask when a control is read. Changing an input intended for Initialize does not prove that an existing solver rebuilt its starting state.

2. Distinguish three kinds of time

Blender frame number, solver-step count and demo warm-up are not one number. Increasing Steps/Frame changes calculations per frame. Run Steps advances animated scenes frame by frame using Steps/Frame; static scenes stay on the current frame. Record which route you used to develop the scene when making repeatable comparisons.

3. Check data before material

If wet sand looks dry, determine whether moisture exists in the data. If a vortex is invisible, check its required material and framing. If a mask fails, inspect its coverage. Changing a material does not fix an empty field, and changing physics does not fix a poorly placed camera.

4. Increase quality in stages

First choose scale, composition and processes on a small grid. Then inspect a representative area at target resolution. Raise rendering cost and recording length afterwards. Accelerated modes may affect accuracy, so make the final judgment using the settings you will actually use.

5. Record comparison conditions

Note add-on and Blender versions, demo or preset, seed, resolution, input files, step count and the changed setting. Preserve camera and lighting when comparing images. Identical seeds do not guarantee bit-identical results across different CPU/GPU backends.

Troubleshooting step by step

Terrain changes or disappears at Initialize

Check the object in Terrain, Current Mesh and Terrain = Sand. A generator prepares a new shape; a loose landscape can be reshaped. Also check whether a thick starting layer simply covers small relief. Return to the pre-initialization copy and make one change.

Painted sand does not appear

Check Starting Sand from a Map and whether Sand Map is empty. An assigned image can take precedence. Confirm you painted the correct grid and are viewing the newly initialized state. Temporarily simplify placement filters to separate brush behavior from regional restrictions.

Terrain is visible but nothing moves

Check Initialize, Play, Freeze Dunes, loose material and wind. For snow or water, check the model's required conditions. Observe a local region for longer: a stable landscape need not change dramatically. Avoid immediately increasing every parameter.

Sand slides off whole mountains or is barely visible

Compare landscape height, width, sand thickness and Dune Scale. Large mountains can dominate dunes. Steep slopes may not retain a thick loose layer. Verify Image Amplitude and Height Scale before changing the physical angle of repose.

A mask has the opposite effect

Read what zero and one mean for the selected field. Erodibility = 0 protects against erosion; another field can interpret zero differently. Check whether you used Clear, Reset or Remove. The first two write values; the last removes the mask. Inspect the preview before the next step.

Material is gray, black or an effect is invisible

Check display mode, material assignment, available files and lighting. Compare Fast Surface or a simple material with the extensive version. If only the complex variant fails, read renderer messages and reduce its cost. A dark surface is not always a solver error.

Blender is too slow

Read actual GPU/CPU status. Reduce grid and material cost, judging the change after one-time preparation completes. Reduce extra effects if they burden the viewport. Solver steps, mesh refresh and rendering have different costs. Run Steps follows animated physical inputs without viewport redraws; optional particle displays refresh at the end. External simulations still need valid Blender caches.

Reopening gives a different result

Check add-on version, external resources and save type. Preserved geometry does not necessarily mean restored solver state. Load the appropriate checkpoint or recorded run if prepared. Verify in a copy before reinitializing an important project.

When reporting a problem, include Blender and DunesForge versions, the demo or a small scene, the sequence of actions, expected behavior and actual result. Include relevant error messages. A small reproducible example helps more than a list of every setting in a large project.

Final project: your own sand-filled valley

This exercise connects the stages you have learned. Aim for a small readable scene that you can reopen, rather than the maximum number of enabled features.

1. Import a gentle heightmap or prepare a regular grid. Preserve it as ground with Current Mesh and Terrain = Sand off.
2. Paint two starting-sand patches: a broad one in the valley and a smaller one beside it. Save before Initialize.
3. Initialize and inspect thickness. Choose one clear wind direction and run short intervals. Stop when the forms work from the camera.
4. Add one obstacle and compare its influence in a copy. Leave rain, snow and vegetation for separate later experiments.
5. Assign a material, adjust detail scale and make a small render. Inspect the area near the obstacle and the edge of the sand.
6. Save the .blend, required checkpoint and a test export. Open a copy and verify restoration.
7. Now add one more element, such as vegetation or a wet shoreline. The preserved simple variant shows what the addition actually changed.

The project is ready for further work when you can identify the terrain source, sand source, active wind, masks and method of preserving the result. Understanding these decisions is more useful than memorizing every option name.

A short checklist before longer work

- I know which object is the terrain and whether its shape should remain the ground.
- I know the starting-sand source: uniform layer, image or painting.

- I inspected the initialized result before starting a long simulation.
- I understand the difference between resolution, area size and dune scale.
- I checked features still active despite being hidden in a simpler UI.
- I saved a working variant before the next experiment.
- I played back a short recording or checked an export's appearance.
- I selected final settings using the actual shot, not only a fast preview.

Find further explanations and individual controls in the complete feature guide. Return with a concrete question; you do not need to learn the whole index before making your first good scene.

Animated objects in Run Steps

Run Steps follows scene animation automatically when the scene contains keyframes, NLA, drivers, constraints, a time-dependent geometry modifier or an animated source-map sequence. Static scenes keep their current timeline frame and use the original fast solver batch.

Example: start at frame 1, set Steps/Frame to 5 and Batch to 100, then press Run Steps. The add-on evaluates frames 2 through 21, simulating five steps at each frame. The timeline stays at frame 21. It continues past the scene End frame instead of looping back. If a batch ends partway through a frame, the next press finishes that frame first.

Assign your object to the appropriate slot and enable that feature: Obstacle for a moving wind barrier, Source for a moving sand emitter, or Imprint Objects for feet, wheels and other contacts. Keyframe the object or its parent, or use drivers or shape keys. Animated batches refresh the object masks every evaluated frame, even when ObsEvery is zero. Imprints also require their Every Frame option. Merely animating an unassigned object does not make it a collider.

The batch avoids viewport redraws. Contact tests may still need intermediate terrain mesh updates, and animated objects cost more to evaluate than a static scene. Do not expect the same speed as an empty static batch. Fast sand and snow emitters, obstacles and object imprints are also sampled between frames, using their actual evaluated Blender animation. This fills a fast emitter's trail

without multiplying its emission: source footprints are averaged over the frame. Obstacles use the space swept during that frame, then return to their final footprint. Prints combine the touching parts of the motion once; an airborne foot still leaves a gap. Playback and Run Steps use the same sampling. Steps/Frame still controls solver steps, not the number of motion samples. Fast motion costs extra object evaluations. Keep a footprint at least one simulation cell wide: subframe sampling cannot recover geometry that the grid cannot resolve. Extreme motion that exceeds the sampling safety limit stops with an explanation instead of silently leaving holes. This samples heightfield inputs; it does not add general 3D object pushing or fluid collisions. External cached simulations must provide valid intermediate evaluations.

Enable Record Evolution and choose Bake Dir before the batch if you want to scrub or render the resulting terrain afterwards. Animated batches record each evaluated frame; a static batch records its final state at the current frame. Moving the timeline backwards does not undo the live simulation: reset and simulate again, or use the recorded bake. External cloth, rigid-body and other cache-driven systems still require a valid Blender cache; Run Steps does not bake those systems for you.

Cloth, Rigid Body and cached Geometry Nodes

First bake the Blender simulation, then run DunesForge. Run Steps reads evaluated geometry from the saved cache; it does not create or repair that cache. A temporary playback preview is not a substitute for a completed bake when the same frames must be read repeatedly.

Save the .blend file before baking. For Cloth, select the simulated object and use Physics Properties > Cloth > Cache. For Rigid Body, use Scene Properties > Rigid Body World > Cache. For a Geometry Nodes simulation zone, use the simulation baking controls for that object. Set the entire frame range you intend to calculate, then Bake. Geometry Nodes must bake the animation range, rather than only one still frame. An Animation Bake node remains recognized when its original animation nodes have been removed; a Still bake alone keeps the static batch path.

Save again after baking. If the bake is stored on disk, keep its folder with the .blend file and use relative paths when moving the project. Reopen the scene and check a few non-consecutive frames before starting a long DunesForge batch. The cached object should still have the same shape and position.

Assign the simulated object to the role you need: Source, Obstacle or Imprint Objects, and enable that role. A simulated object elsewhere in the scene does not automatically affect the dunes. Geometry Nodes can output a mesh, mesh instances, nested mesh instances or a mixture of mesh and instances. DunesForge reads those instances without adding a Realize Instances node or editing the node graph. Point-only or volume-only output needs a suitable surface before it can act as a heightfield object.

Return to the first intended frame, Reset DunesForge if this is a new run, and Initialize. Choose Steps/Frame and Run Steps. For example, 70 steps with Steps/Frame = 5 advances from frame 1 to frame 15, so the Blender bake must cover that interval. Run Steps can continue past the scene End marker; that does not extend an external bake. Stay inside its saved range.

Changing the cloth setup, rigid-body simulation or node simulation may require deleting and rebuilding its Blender bake. After such a change, reset the dune simulation before comparing results. Use the same Blender version to generate and consume a Geometry Nodes bake: Blender does not guarantee that its cache format is compatible between versions.

This workflow reads cached objects as heightfield inputs. It does not turn DunesForge into a general two-way Cloth/Rigid Body coupling system. A cloth reacting to dunes that themselves depend on that cloth forms a feedback loop and needs a separately designed simulation workflow.

Blender: Cloth Cache | Blender: Rigid Body World | Blender: Geometry Nodes Baking

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Stopping, continuing and recording a long batch

Run Steps started from the panel can be stopped with Esc. It finishes the current computation chunk first, then keeps the terrain, step count and current animation frame. Waiting for a dense object mask or a GPU calculation can still take time; Esc is handled between chunks, rather than inside a GPU kernel.

The Batch field changes to the number of steps still missing from the interrupted request. Press Run Steps again to complete those steps. For example, stopping a 1,000-step request after 275 leaves Batch = 725. You can also enter a different number if you want to continue for longer or

shorter. Do not Reset or Initialize between stopping and continuing: those actions start a new simulation.

If you stop partway through a frame, DunesForge keeps the remaining Steps/Frame work for that frame. Continuing finishes it before advancing the animation. Its object motion sampling is reused and an every-frame imprint is not stamped again. Leave Steps/Frame, scene inputs, object animation and timeline position unchanged when you want the same result as an uninterrupted batch. Editing these makes a different experiment.

For a recorded animation, enable Record Evolution, choose Bake Dir and use a separate Run ID before starting. Animated batches write each evaluated timeline frame; a partly simulated frame is updated when its remaining steps finish. A static scene records the latest state at its current frame, rather than inventing an animated timeline. Check a short recording in the bake viewer before committing to a long run. With final-quality float32 recording, a 1024 × 1024 terrain uses about 4 MiB per frame, or about 4 GiB for 1,000 frames, before any additional masks. Default float16 uses about half as much space.

A recorded terrain animation is for replay and rendering. It is not a complete checkpoint of every internal solver field. The Esc-and-continue workflow keeps the live solver in the current Blender session; saving a .blend or a terrain recording does not promise the same continuation after restarting Blender. Save the project, required object caches and recording folders as separate parts of your workflow.

For dense scenes, first run a representative short interval with the same objects and settings. Increase simulation resolution only when the shot needs it. Display Resolution reduces the viewport mesh; it does not reduce the simulation grid or the recorder output. Large object collections and fast motion also add mask evaluations. Verify free disk space before a long recording. A failed frame replacement preserves the previously valid heightfield file and reports the failure in the log; it does not make an incomplete recording complete.

Python calls using EXEC_DEFAULT remain synchronous for scripts and background processing. The Esc gesture applies to a batch invoked from the interactive panel. It is an orderly stop in a running Blender session, not recovery from a process crash, device loss or a partially failed numerical step. Check the error log and reinitialize after a numerical or GPU failure instead of assuming that an interrupted calculation is a valid checkpoint.